

An Analysis of Semantic segmentation

Semantic segmentation

Semantic segmentation -- Part 1

Fast marching methods The fast marching method has been used in image segmentation, and this model has been improved (permitting both positive and negative propagation speeds) in an approach called the generalized fast marching method.

Semantic segmentation -- Part 2

The Laplacian operator is employed such that the partial derivatives are derived from a specific equation. The second partial derivative of $f(x, y)$ with respect to x and y are given by:

Semantic segmentation -- Part 3

Here $\lambda \in \Lambda$, the set of all possible labels. 3. M step: The established relevance of a given feature set to a labeling scheme is now used to compute the a priori estimate of a given label in the second part of the algorithm. Since the actual number of total labels is unknown (from a training data set), a hidden estimate of the number of labels given by the user is utilized in computations.

Semantic segmentation -- Part 4

This above equation is used to determine whether a point in the image is an isolated point based on the response magnitude $|R(x, y)|$ and a threshold value T . If the response magnitude is greater than or equal to the threshold, the function returns 1, indicating the presence of an isolated point; otherwise, it returns 0. This helps in the effective detection and segmentation of isolated points in the image.

Semantic segmentation -- Part 5

Graph partitioning methods Graph partitioning methods are an effective tools for image segmentation since they model the impact of pixel neighborhoods on a given cluster of pixels or pixel, under the assumption of homogeneity in images. In these methods, the image is modeled as a weighted, undirected graph. Usually a pixel or a group of pixels are associated with nodes and edge weights define the (dis)similarity between the neighborhood pixels. The graph (image) is then partitioned according to a criterion designed to model "good" clusters. Each partition of the nodes (pixels) output from these algorithms are considered an object segment in the image; see Segmentation-based object categorization. Some popular algorithms of this category are normalized cuts, random walker, minimum cut, isoperimetric partitioning, minimum spanning tree-based segmentation, and segmentation-based object categorization.

Semantic segmentation -- Part 6

$$P(\lambda) = \frac{\sum_{\lambda \in \Lambda} P(\lambda | f_i) | \Omega |}{\sum_{\lambda \in \Lambda} P(\lambda | f_i) | \Omega |}$$

Semantic segmentation -- Part 7

Dual clustering method This method is a combination of three characteristics of the image: partition of the image based on histogram analysis is checked by high compactness of the clusters (objects), and high gradients of their borders. For that purpose two spaces have to be introduced: one space is the one-dimensional histogram of brightness $H = H(B)$; the second space is the dual 3-dimensional space of the original image itself $B = B(x, y)$. The first space allows to measure how compactly the brightness of the image is distributed by calculating a minimal clustering k_{min} . Threshold brightness T corresponding to k_{min} defines the binary (black-and-white) image – bitmap $b = \phi(x, y)$, where $\phi(x, y) = 0$, if $B(x, y) < T$, and $\phi(x, y) = 1$, if $B(x, y) \geq T$. The bitmap b is an object in dual space. On that bitmap a measure has to be defined reflecting how compact distributed black (or white) pixels are. So, the goal is to find objects with good borders. For all T the measure $MDC = G/(k \times L)$ has to be calculated (where k is difference in brightness between the object and the background, L is length of all borders, and G is mean gradient on the borders). Maximum of MDC defines the segmentation.

Semantic segmentation -- Part 8

Application of Isolated Point Detection in X-ray Image Processing The detection of isolated points has significant applications in various fields, including X-ray image processing. For instance, an original X-ray image of a turbine blade can be examined pixel-by-pixel to detect porosity in the upper-right quadrant of the blade. The result of applying an edge detector's response to this X-ray image can be approximated. This demonstrates the segmentation of isolated points in an image with the aid of single-pixel probes.

Semantic segmentation -- Part 9

Alternative algorithms A range of other methods exist for solving simple as well as higher order MRFs. They include Maximization of Posterior Marginal, Multi-scale MAP estimation, Multiple Resolution segmentation and more. Apart from likelihood estimates, graph-cut using maximum flow and other highly constrained graph based methods exist for solving MRFs.

Semantic segmentation -- Part 10

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Semi-automatic segmentation In one kind of segmentation, the user outlines the region of interest with the mouse clicks and algorithms are applied so that the path that best fits the edge of the image is shown. Techniques like SIOX, Livewire, Intelligent Scissors or IT-SNAPS are used in this kind of segmentation. In an alternative kind of semi-automatic segmentation, the algorithms return a spatial-taxon (i.e. foreground, object-group, object or object-part) selected by the user or designated via prior probabilities.

Semantic segmentation -- Part 11

Groups of image segmentation Semantic segmentation is an approach detecting, for every pixel, the belonging class. For example, in a figure with many people, all the pixels belonging to persons will have the same class id and the pixels in the background will be classified as background. Instance segmentation is an approach that identifies, for every pixel, the specific belonging instance of the object. It detects each distinct object of interest in the image. For example, when each person in a figure is segmented as an individual object. Panoptic segmentation combines both semantic and instance segmentation. Like semantic segmentation, panoptic segmentation is an approach that identifies, for every pixel, the belonging class. Moreover, like in instance segmentation, panoptic segmentation distinguishes different instances of the same class.

Semantic segmentation -- Part 12

Parametric methods Lagrangian techniques are based on parameterizing the contour according to some sampling strategy and then evolving each element according to image and internal terms. Such techniques are fast and efficient, however the original "purely parametric" formulation (due to Kass, Witkin and Terzopoulos in 1987 and known as "snakes"), is generally criticized for its limitations regarding the choice of sampling strategy, the internal geometric properties of the curve, topology changes (curve splitting and merging), addressing problems in higher dimensions, etc.. Nowadays, efficient "discretized" formulations have been developed to address these limitations while maintaining high efficiency. In both cases, energy minimization is generally conducted using a steepest-gradient descent, whereby derivatives are computed using, e.g., finite differences.